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Increased Metal Concentrations in Giant Sungazer Lizards (*Smaug giganteus*) from Mining Areas in South Africa

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Abstract Environmental contaminants from anthropogenic activity such as mining can have profound health effects on the animals living in adjacent areas. We investigated whether inorganic contaminants associated with gold-mining waste discharges were accumulated by a threatened species of lizard, *Smaug giganteus*, in South Africa. Lizards were sampled from two mining sites and two control sites. Blood samples from the most contaminated mining site had significantly greater concentrations of lithium, sodium, aluminum, sulfur, silicon, chromium, manganese, iron, nickel, copper, tungsten, and bismuth than the remaining sites. Contaminant concentrations were not significantly related to lizard body condition, although these relationships were consistently negative. The adult sex ratio of the population inhabiting the most contaminated site also deviated from an expected 1:1 ratio in favour of female lizards. We demonstrate that lizards at these mining sites contained high concentrations of heavy metals that may be imposing as yet poorly understood costs to these lizards.

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The impacts of underground mining on the environment are varied but primarily include physical changes to the structure and stability of mined areas (Altun et al. 2010), and pollution (Nagajyoti et al. 2010). Gold mining, in particular, is often the cause of metal pollution in various environments and can result in serious risks to environmental and human health (e.g., Basu et al. 2011; Picado et al. 2010; Wu et al. 2011). South Africa is well known for its gold and uranium mining industries; these industries are often the source of metal contaminants, such as mercury (Hg), in the surrounding environments (Walters et al. 2011). Poorly established precautionary and rehabilitation measures around many mines and tailings dams can pollute soils and groundwater with various inorganic contaminants (Aucamp and van Schalkwyk 2003; Rösner and van Schalkwyk 2000).

Both organic and inorganic contaminants used during mining processes can be accumulated by organisms inhabiting these environments (e.g., Parker 2004; van Eeden 2003) and influence their physiology and fitness. For example, ingestion of metal contaminants in fish (*Oncorhynchus mykiss*) decreases growth (Hansen et al. 2004), whereas exposure to increased copper (Cu) concentrations in water was associated with immune system deficiencies in the same species (Dethloff and Bailey 1998). In addition, birds (*Parus major*) sampled at a site with increased concentrations of mainly lead (Pb) and cadmium (Cd) near a metallurgic smelter had a decreased immune response (Snoeijs et al. 2004). Increases in territorial aggressive behaviour and decreases in egg-hatching success from great tit pairs exposed to high environmental concentrations of arsenic (As), Cd, Cu, Pb, and zinc (Zn) have also been reported (Janssens et al. 2003) as has damage to the DNA structure of fish (*Sparus aurata*) and molluscs (*Scapharca inaequivalvis*) due to exposure to increased concentrations (0.1 ppm) of Cu sulfate (Gabbianelli et al. 2003).

Exposure to metal contaminants also potentially affects organisms at a population level. For example, skewed sex ratios in favour of adult female birds have been reported in some populations of white-tailed ptarmigan (*Lagopus leucurus*) exposed to increased concentrations of Cd in the Colorado Rocky Mountains ore belt (Larison et al. 2000). Recent genetic analyses also showed population-level genetic effects on fish populations exposed to chronic metal pollution (e.g., Durrant et al. 2011; Pierron et al. 2011).

Reptiles appear to be decreasing on a global scale mainly due to habitat loss, competition, and predation from invasive species, diseases, and parasitism; unsustainable use (overcollection for food, pet trade, and biological supply houses); climate change; and environmental pollution (e.g., Araújo et al. 2006; Poppy et al. 2000; Reading et al. 2010). Reptiles (lizards in particular) are considered useful bioindicator organisms because they are mostly insectivorous (Bishop and Gendron 1998), are integral to many food webs (Campbell and Campbell 2000; Lambert 1997), and show high site fidelity (Burger et al. 2004; Lambert 1993). These characteristics of lizards provide opportunities to compare contaminant effects on individuals and populations inhabiting contaminated and reference sites within small geographical areas (Hopkins 2000). Despite these features, reptiles remain the least studied vertebrate group regarding the accumulation and effects of inorganic contaminants (Campbell and Campbell 2002; Hopkins 2000). The few studies on potential effects of inorganic pollution on reptiles have generally been in controlled environments, such as laboratories (e.g., Brasfield et al. 2004; Hopkins et al. 2001, 2002, 2005). Both laboratory and field studies mostly obtained the relevant inorganic uptake concentration through destructive sampling of tissues from major organs, such as kidneys and livers (e.g., Anan et al. 2001; Burger et al. 2004; Loumbourdis 1997 [but see Jeffree et al. 2001]).

The giant girdled or sungazer lizard (*S. giganteus* [formerly *Cordylus giganteus*; Stanley et al. 2011]) is endemic to the highveld region of South Africa (Fig. 1). The highveld supports extensive farming activities (cropping) as well as gold- and coal-mining. Sungazer lizards are threatened, and the species is listed as vulnerable in the IUCN Red List of Threatened Species (http://www.redlist.org). Until now, the primary threats to this species have been direct loss of habitat through agriculture and industrial development as well as illegal collection for the pet and *muti* (traditional medicine) trades (Whiting et al. 2011). However, large areas of the Gauteng and Free State Provinces (the latter is where most *S. giganteus* populations occur) have reportedly been affected by contaminated dust, seepage, and groundwater emanating from gold-mining activities (Coetzee 1995; Rösner and van Schalkwyk 2000; Weiersbye et al. 2003). These contaminants may provide an additional threat to populations of *S. giganteus* if the lizards are negatively affected by exposure to these contaminants.

Our primary aim was to establish whether populations of *S. giganteus* occurring in mining areas had increased levels of inorganic contaminants compared with those in control sites. We also tested the prediction that lizards from mining areas would be in poorer body condition and tested whether population sex ratios deviated from a 1:1 ratio in both mining and control sites.

Materials and Methods

Study Area

We sampled lizards from four sites in the Free State Province (Fig. 1). Mining site 1, in the Welkom area of the Free State Province, contains a natural saltpan. The pan has been used during a period of 40 years for the legally licensed discharge and evaporation of gold mine process water and, to a lesser extent, purified sewage effluent (Weiersbye et al. 2003). The pan water and sediment, as well as the surrounding soils, are contaminated with a variety of elements. These elements are associated with acid mine drainage and mine process water discharge and include sulfur, chlorine, sodium, calcium (Ca), magnesium (Mg), potassium (K), iron (Fe), aluminum (Al), manganese (Mn), strontium (Sr), titanium (Ti), Zn, chromium (Cr), cobalt, nickel (Ni), uranium, gold, Hg, Cu, As, lithium (Li), vanadium (V), yttrium, and selenium (Weiersbye et al. 2003).

Mining site 2 (Slimes dam site) is approximately 10 km from mining site 1, and the surface water streams and soils are contaminated by seepage from adjacent slimes dams (I. M. Weiersbye [unpublished data]). At the time of our study, a small population (approximately <10 individuals) of *S. giganteus* appeared to be persisting on this site. We selected two control sites. Control site 1 was situated on undisturbed rangeland near Lindley in the Free State Province. The population of lizards on this farm was fenced off and protected from most anthropogenic influences. Control site 1 was free of any cattle, sheep, or wild antelope grazing, and the grassland was therefore intact. Control site 2 was an overgrazed game ranch situated approximately 15 km northeast of Welkom. No mining activity has taken place in its immediate vicinity, and it is believed to be free of any contaminants associated with mining; however, the possibility of wind-blown contaminants could not be excluded.

Sungazer lizards are thought to be sensitive to changes in general habitat, such as vegetation type (Jacobsen et al. 1990). Vegetation cover can influence prey availability (see

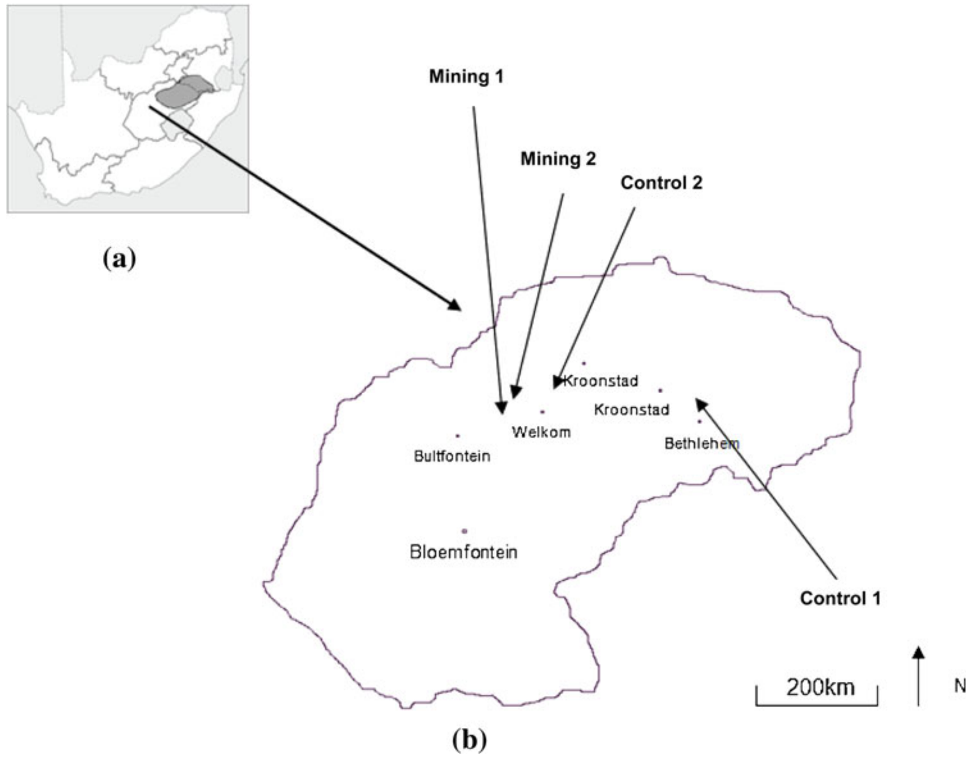

Fig. 1 **a** Map of South Africa with provincial boundaries showing the geographical distribution of *S. giganteus* (*dark shading*) based on de Waal (1978), Jacobsen et al. (1990), and Ruddock (2000). **b** Enlarged map of the Free State Province showing the approximate locality of the four study sites. The exact locations are not shown, in accordance with IUCN guidelines, to prevent illegal collecting

later text) and thus the physiological condition of lizards. We therefore estimated the percentage of vegetation cover for each site under investigation. We did this by visually estimating the percentage of ground covered by vegetation (aerial cover) in 10 1 × 1-m quadrants for each site. An overhead digital photograph was also taken at head height over each quadrant to obtain a visual archival record.

Lizard Sampling

We obtained tissue and blood samples from adult lizards from mining site 1 and control site 2 during February 2004 and September to October 2004. Lizards from mining site 2 were sampled only during February 2004, and the population of control site 1 was sampled during September and October 2004. To prevent resampling of the same individual, all individual lizards were marked using passive integrated transponders (PIT tags; Identipet) with a unique alpha-numeric code. PITs were injected subcutaneously into the postero-femoral region of the hind legs.

We weighed lizards to the nearest 0.1 g on a digital scale, measured snout–vent length (SVL) and tail length to the nearest 1 mm with a ruler and head length, width, and height to the nearest 0.1 mm using digital callipers. We sexed lizards by checking for the presence of generation glands in the femoral and forearm regions. These glands are only present in male lizards (van Wyk 1992). We took a blood and/or tissue sample (see later text) before releasing lizards within 1 day at the point of capture.

Tissue Collection

Because *S. giganteus* is a species of conservation concern, we used nonlethal techniques for obtaining tissue and blood samples. The levels of elements in blood and tissue were expected to differ due to the half-life of each contaminant and the affinity of particular cell types for different ions (Clemans et al. 2002). Blood sampling holds obvious advantages compared with tissue sampling because it is less invasive and repeatable on the same individual over time. However, tissue samples from tail clippings in reptiles may be more reliable than blood for quantifying the accumulation of contaminants because of the composite of blood, skin, bone, and muscle (Jackson et al. 2003). We sampled both tissue and blood to ensure that the accumulated contaminants were detected and to compare tail tissue and blood as indicators of contaminant levels. We removed 5–10 mm of tissue from the tips of *S. giganteus* tails (Hopkins et al. 2001; Jackson et al. 2003) for analysis of inorganic contaminants. We also obtained a 150-µl blood sample from the suborbital sinus of each lizard using two 75-µl heparinised microcapillary tubes. This procedure is a standard method for obtaining blood from free-ranging lizards (Mautz and Nagy 2000; Nagy 1993; Znari and Nagy 1997). Each blood sample was stored in a polypropylene container, centrifuged for 7 min at 3,000 rpm, and the plasma removed using a micropipette. After removal, the plasma and red blood cells were frozen and sent to the laboratory for further analysis.

Elemental Analyses

We quantified elemental concentrations between lizards from the mining-affected sites and lizards from areas not affected by mining activity (control sites 1 and 2) to determine metal uptake in lizards from mining-contaminated areas. Inorganic concentrations were quantified for the blood and tissue samples in the laboratory. We cleaned all tail tissue (not red blood or plasma samples) thoroughly with toluene before analysis to remove external soil and vegetative particles. All samples analyzed were dried and then underwent closed microwave digestion in aqua regia (55 % solution of nitric acid, HNO_3, and 32 % solution of hydrochloric acid, HCl) using a Multiwave 3000 microwave (Anton Paar). The digested solutions were cooled to room temperature and then diluted to volume in 50 ml polyvinylpyrrolidone volumetric flasks using Milipore double-distilled water. The concentrations of 21 elements were analyzed in triplicate using inductively coupled plasma-optical emission spectrometry on a Ciros-CCD analyzer (Spectro). Plasma power was set at 1300 W; coolant flow at 13 l/min; auxiliary flow at 1 l/min; and nebulizer flow at 1 l/min. Blank samples were run on for all digestions, and background correction was applied.

We initially compared concentrations of 19 elements found in whole blood and tissue between lizard sexes. Concentrations of elements in whole blood were calculated by combining data from plasma and red blood samples for individual animals (and accounting for the weights of each plasma sample relative to its associated red blood sample). Four elements (Ti, barium, Hg, and Pb) were lower than minimum detection limits for most of the samples analyzed. None of the elements occurred in significantly different concentrations (Student *t* tests; $p > 0.05$) between sexes. Because no differences were observed in elemental concentrations between sexes, we subsequently combined samples from both sexes for all further comparisons between tissue types and populations.

Body Condition, Prey Availability, and Sex Ratios

We performed a regression of body weight (log-transformed) on SVL (log-transformed) to obtain an index of body condition (Dunlap and Mathies 1993; Jakob et al. 1996; Cuadrado 1998; Anderholm et al. 2004), for comparison among sites. Because lizards were sampled from the control site only during the final sampling period, we restricted the comparison only to adult lizards caught during this period, on first capture, from all sites.

We used two-factor analysis of variance (ANOVA) to assess differences between sites and sexes in body condition. Because food availability directly affects body condition, we measured prey abundance at the three main sites under investigation (mining site 1, control site 1, and control site 2). Mining site 2 was excluded because of a small sample size ($n = 7$). *S. giganteus* are insectivores (van Wyk 2000); thus, invertebrate abundance was used as an indicator of relative prey availability. We used a cross array of pitfall traps (Perner and Schueler 2004) to sample invertebrate abundance by burying 25 unbaited plastic bowls (diameter 200 mm; depth 150 mm) with their open ends flush with the ground. Individual traps were arranged in a cross with distances of traps from the centre in the following increasing order: 0.5, 3, 6, 12, 24, and 48 m. Each trap was covered with a metal grid consisting of squares small enough to prevent the entry of any small vertebrates. Small amounts of automotive antifreeze were used to kill and preserve all invertebrates caught in the traps. We set 25 traps at each site for a total period of 4 weeks during summer. We checked each trap once a week and collected and preserved all trapped insects for later identification in the laboratory. All invertebrates were sorted, identified to the lowest taxon possible (mostly family), and counted.

Data Analysis

Statistical analyses were performed in R (R 2.11.1, R Development Core Team 2008). Data are reported as means ± SDs. Statistical significance was set at $\alpha = 0.05$.

Results

Elemental Concentrations

We found significant differences between sites in concentrations of the following elements in *S. giganteus* whole-blood samples: Na, Ni, S, Al, Cr, Fe, Mn, and silicon (Si) (Fig. 2). We plotted the sample values for all of the elements where we found a significant site difference. When ranked from highest to lowest, mining site 1 samples always appeared among the top three samples, but their order was not consistent (i.e., the significance is not due to one or two samples being consistently greater). Therefore, the elemental concentrations of individual samples did not covary within study sites. Post hoc tests showed that these significant differences were always between concentrations measured in blood samples from mining site 1 and the other three sites (Fig. 2). Differences in concentrations of Bi, Li, Cu, and W were also apparent between mining site 1 and the remaining sites (Fig. 2), although these were not significant. Whole-blood concentrations of K, P, Mg, Ca, and Zn did not differ significantly between sites. Elemental concentrations found in tail tissue from lizards did not differ significantly between sites. Measured elemental concentrations for lizard tail tissue are listed in Table 1.

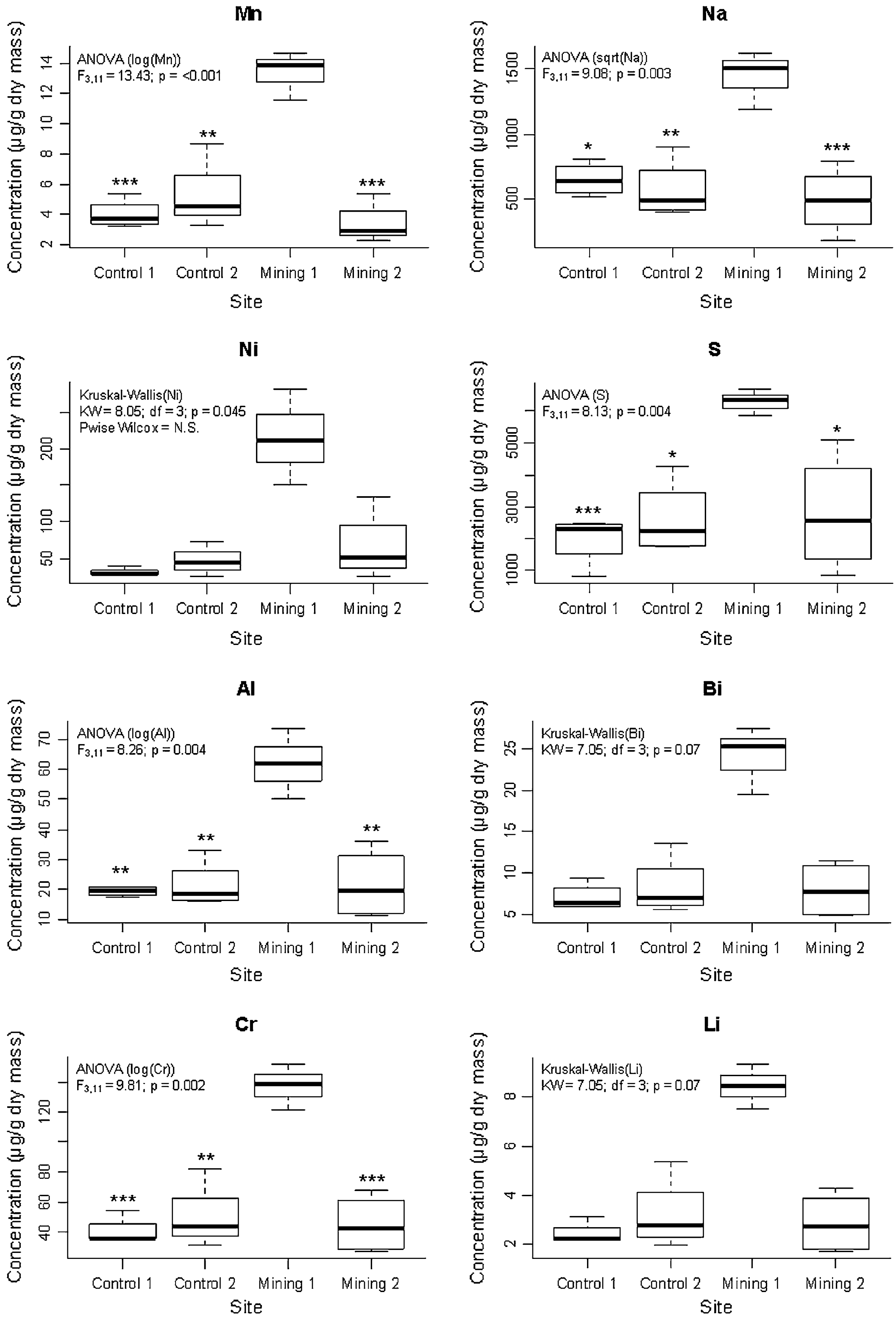

Fig. 2 Box plots showing the measured elemental concentrations in blood samples between sites: mining site 1 ($n = 3$), mining site 2 ($n = 4$), control site 1 ($n = 4$), and control site 2 ($n = 4$). Statistical test outputs testing for differences in concentrations between sites are provided in each plot, and significant differences, compared with mining site 1 are indicated on the plots (***$p < 0.005$; **$p < 0.01$; *$p < 0.05$). *Bold line* median; box = 25th and 75th percentiles; *whiskers* maximum values

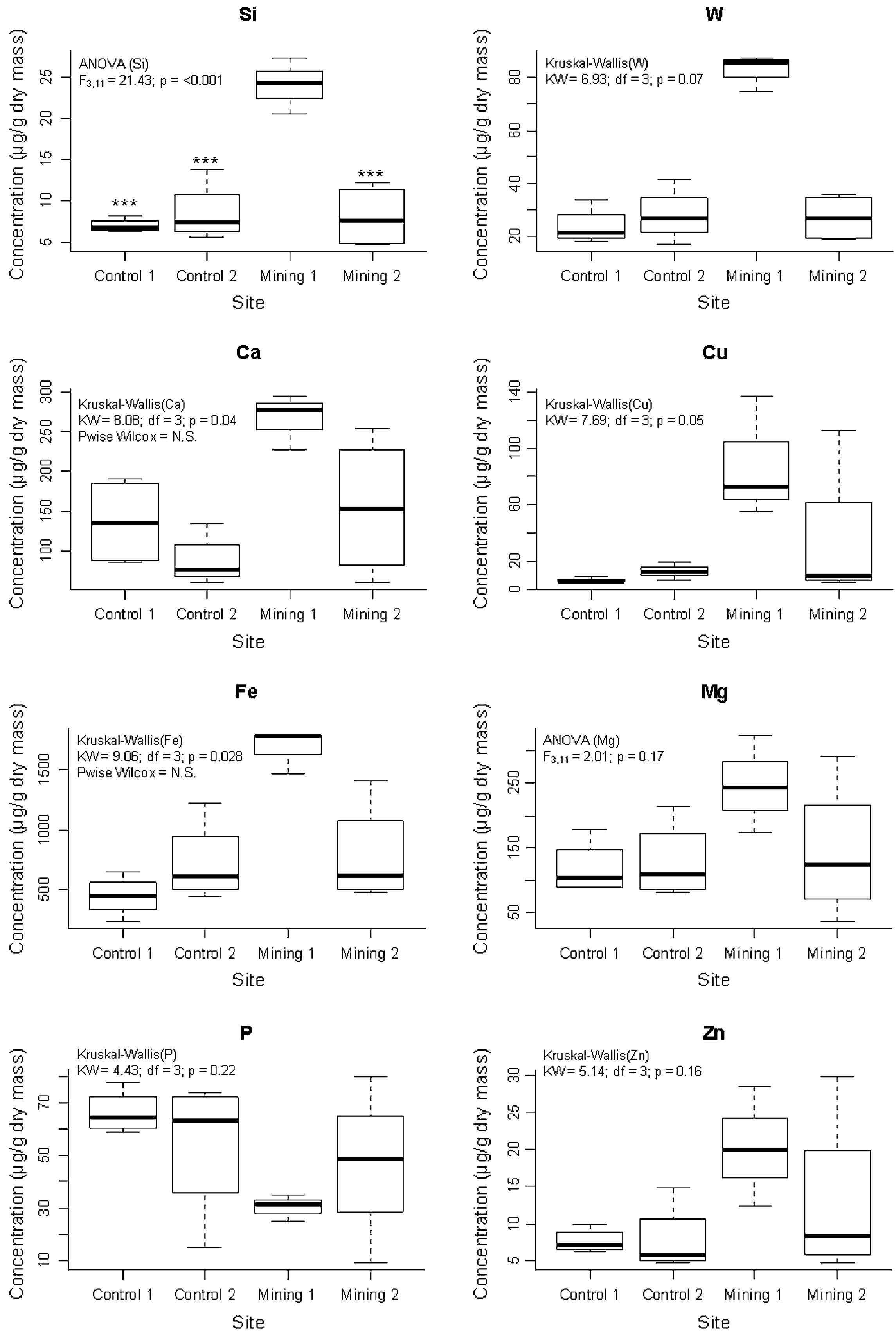

Fig. 2 continued

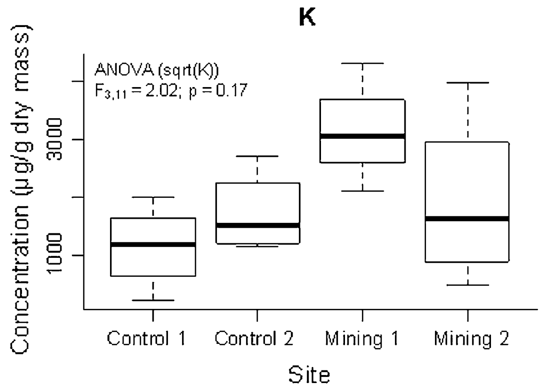

Fig. 2 continued

Body Condition, Prey Availability, and Sex Ratios

Body condition was not significantly different between the sexes ($F_{1,109} = 0.23$; $p = 0.64$), but was significantly different between sampling sites ($F_{2,109} = 30.03$; $p < 0.001$) (Fig. 3). Lizards from mining site 1 and control site 2 were in significantly poorer condition than lizards from the control site 1 population (post hoc Tukey test: mining site 1—control site 1, $p < 0.001$; control site 2—control site 1, $p < 0.001$). We did not detect any significant differences in condition between lizards from mining site 1 and control site 2.

A series of linear regression models relating body condition with tail tissue elemental concentrations indicated mostly nonsignificant relationships between elemental concentrations and body condition (Table 2). Ca concentration displayed a significant positive relationship with body condition in the sampled lizards. Body condition showed a negative relationship with a number of metal concentrations (notably Al, Cu, Fe, Li, Mn, and Ni) measured in tail tissue, although these were not statistically significant (Fig. 4; Table 2). All elements measured in blood samples showed negative but statistically nonsignificant relationships with body condition (Table 2).

The total numbers of invertebrates sampled from the relevant orders and families were used as an index of prey availability. Prey availability was significantly different between sites ($\chi^2 = 29.589$; $df = 2$; $p = <0.001$). Invertebrate abundance was highest at mining site 1. Both control sites had similar abundance but showed significantly less invertebrate abundance than mining site 1.

Neither control site 2 nor control site 1 lizard populations deviated significantly from a 1:1 sex ratio (control site 2: $\chi^2 = 0.56$; $df = 1$; $p = 0.76$; control site 1: $\chi^2 = 0.25$; $df = 1$; $p = 0.88$). However, mining site 1 showed a

Table 1 Mean concentrations (μg/g dry mass), SDs, and test statistics of all elements sampled in *S. giganteus* tail tissue between sites

Element	Mining site 1 ($n = 12$)	Mining site 2 ($n = 4$)	Control site 1 ($n = 4$)	Control site 2 ($n = 4$)	Test type	Test statistics	*df*	*p*
Li	2.5 ± 1.5	2.5 ± 0.3	1.8 ± 0.1	3 ± 0.9	K–W	9.7	3	0.02
Na	703.4 ± 318.5	562.4 ± 43.5	643.8 ± 94.2	747.7 ± 199	K–W	4.06	3	0.26
Mg	2392.9 ± 507.6	2105.9 ± 151.4	2618.3 ± 393.3	2351.1 ± 268.8	K–W	5	3	0.17
Al	167.2 ± 152.1	220.7 ± 112.9	135.4 ± 16.3	303.2 ± 92.9	ANOVA (log)	1.99	3,20	0.15
Si	22.7 ± 19	25 ± 10.4	30.1 ± 6.4	74.4 ± 67.7	ANOVA (log)	3.36	3	0.04
P	11.8 ± 18.2	12.3 ± 10.3	15.1 ± 6.8	58.3 ± 63.5	ANOVA (log)	1.804	3	0.165
S	8031.2 ± 3804.7	6366.5 ± 789.4	10397.6 ± 4603.1	77900 ± 4055.9	K–W	2.26	3	0.52
K	927.6 ± 640.3	579.9 ± 229.7	411.2 ± 140.8	538.1 ± 72.8	ANOVA (log)	2.85	3,20	0.06
Ca	118530.4 ± 21598.1	110850.1 ± 5773.6	133002.8 ± 22479.5	120916.3 ± 20281.6	ANOVA (log)	0.85	3,20	0.49
Ti	21.3 ± 19.6	26.1 ± 10.1	16.7 ± 0.5	28.7 ± 19.7	ANOVA (log)	0.4	3,20	0.76
Cr	32.1 ± 20.5	26.3 ± 0.7	26 ± 0.7	34.3 ± 14.5	K–W	5.42	3	0.14
Mn	11.3 ± 4.9	13.7 ± 5.8	8.7 ± 1.2	16.5 ± 7.8	ANOVA (log)	2.07	3,20	0.14
Fe	406.2 ± 404.1	452.7 ± 183.4	264.2 ± 21.1	547.4 ± 212.6	ANOVA (log)	1.48	3,20	0.25
Ni	41 ± 57.5	23.1 ± 2.5	20.6 ± 1.7	26.7 ± 14.8	K–W	7.45	3	0.06
Cu	18.7 ± 35.7	7.3 ± 2.3	5.8 ± 1.8	7.4 ± 4.1	K–W	5.58	3	0.13
Zn	179.6 ± 25.8	150.4 ± 27	147.5 ± 6	163.1 ± 13.6	ANOVA	3.02	3,20	0.05
W	20.6 ± 17.6	14.7 ± 3.5	18.7 ± 2	19.2 ± 9.2	K–W	4.26	3	0.24
Bi	23.8 ± 20.8	26.8 ± 11.6	32.2 ± 7.3	79.8 ± 71.9	ANOVA (log)	3.46	3,20	0.04

We used K–W in lieu of analysis of variance when assumptions of normality could not be met. Significant differences were found in Li, Si, and Bi between sites

K–W Kruskal–Wallis; *ANOVA* analysis of variance

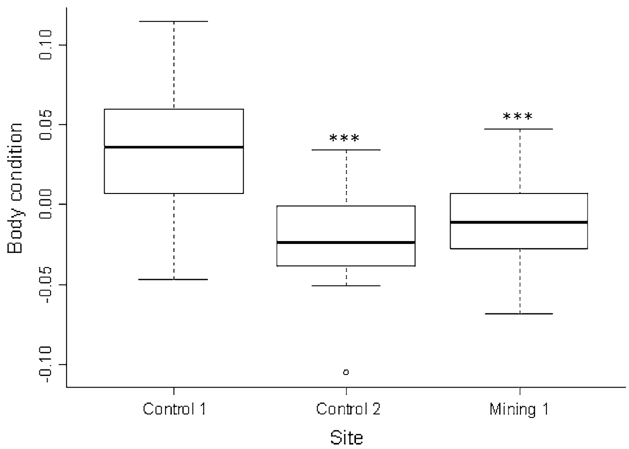

Fig. 3 Box plots of mean residual values (body condition) obtained after regressing log-transformed body weight over log-transformed SVL of *S. giganteus* from three sites. Means (±SDs) values are reported. Statistically significant differences are indicated compared with control site 1 (***$p < 0.005$)

significant deviation from a 1:1 ratio and had significantly ($\chi^2 = 7.54$; $df = 1$; $p = 0.02$) more female (71 %) than male lizards (29 %).

Discussion

Elemental Concentrations in *S. giganteus* Tissue

Our results indicate the uptake of a number of inorganic contaminants by *S. giganteus* in areas affected by mining effluent in the Free State Province of South Africa. Blood samples obtained from lizards at mining site 1, an evaporation pan, contained increased concentrations (when compared with samples obtained from other sites) of Na, Ni, S, Cu, Bi, Al, Cr, Fe, Mn, Si, Li, and tungsten. Zn concentrations were evidently also increased in blood samples from lizards at mining site 1, although statistical significance was likely prevented by the large variation in Zn concentrations measured from mining site 2 (Fig. 2). Some of these elements had previously been reported as being increased in the water and/or sediments of mining site 1 site (e.g., Li, Na, Al, Cr, Mn, Fe, and Ni [Weiersbye et al. 2003]). Although we did not measure metal accumulation in organ tissues of lizards (i.e., liver, kidneys, and gonads), the high metal concentrations in blood suggest that such organs were also likely to contain high concentrations of heavy metals. Concentrations of metals in tail tissue were mostly not significantly different between sampling sites, with the exceptions being Li, Si, and Bi, which were greater in tail tissue from animals from control site 1 compared with the other sampling site.

Table 2 Linear regression outputs describing relationships between element concentrations in tail tissue as well as blood with body condition of sampled lizards

Element	Linear model equation	R^2	F	p
K_{tail}	$y = -0.001x + 0.749$	0.125	$3.13_{1,22}$	0.091
K_{blood}	$y = -0.0x + 1.005$	0.153	$2.351_{1,13}$	0.149
Na_{tail}	$y = 0.001x - 0.342$	0.043	$0.995_{1,22}$	0.329
Na_{blood}	$y = -0.0 + 0.576$	0.011	$0.146_{1,13}$	0.709
Ni_{tail}	$y = -0.042x + 1.286$	0.065	$1.449_{1,22}$	0.242
Ni_{blood}	$y = -0.002x + 0.581$	0.035	$0.469_{1,13}$	0.505
S_{tail}	$y = 0.0x + 0.098$	0.004	$0.094_{1,22}$	0.763
S_{blood}	$y = -0.0x + 0.0$	0.033	$0.437_{1,13}$	0.52
Mg_{tail}	$y = 0.001x - 1.262$	0.071	$1.681_{1,22}$	0.208
Mg_{blood}	$y = -0.005x + 1.081$	0.15	$2.29_{1,13}$	0.154
Mn_{tail}	$y = -0.038x + 0.704$	0.043	$0.98_{1,22}$	0.333
Mn_{blood}	$y = -0.015x + 0.481$	0.991	$0.051_{1,13}$	0.825
Si_{tail}	$y = 0.001x + 0.223$	0.000	$0.009_{1,22}$	0.925
Si_{blood}	$y = -0.015x + 0.559$	0.013	$0.167_{1,13}$	0.7
Li_{tail}	$y = -0.21x + 0.76$	0.06	$1.398_{1,22}$	0.25
Li_{blood}	$y = -0.055x + 0.612$	0.021	$0.279_{1,13}$	0.606
W_{tail}	$y = -0.009x + 0.408$	0.012	$0.274_{1,22}$	0.606
W_{blood}	$y = -0.006x + 0.604$	0.02	$0.27_{1,13}$	0.612
Zn_{tail}	$y = 0.001x + 0.019$	0.001	$0.025_{1,22}$	0.875
Zn_{blood}	$y = -0.037x + 0.818$	0.1	$1.459_{1,13}$	0.249
Ca_{tail}	$y = 0.0x - 3.169$	0.313	$10.03_{1,22}$	0.005
Ca_{blood}	$y = -0.002 + 0.73$	0.036	$0.475_{1,13}$	0.503
Cu_{tail}	$y = -0.126x + 1.224$	0.098	$2.287_{1,22}$	0.145
Cu_{blood}	$y = -0.005x + 0.558$	0.053	$0.73_{1,13}$	0.408
Bi_{tail}	$y = 0.001x + 0.225$	0.000	$0.008_{1,22}$	0.931
Bi_{blood}	$y = -0.014x + 0.548$	0.011	$0.15_{1,13}$	0.705
Al_{tail}	$y = -0.002x + 0.536$	0.039	$0.884_{1,22}$	0.357
Al_{blood}	$y = -0.004 + 0.5$	0.005	$0.066_{1,13}$	0.802
Cr_{tail}	$y = -0.009x + 0.512$	0.018	$0.412_{1,22}$	0.528
Cr_{blood}	$y = -0.003x + 0.585$	0.017	$0.222_{1,13}$	0.645
Fe_{tail}	$y = -0.001x + 0.562$	0.056	$1.308_{1,22}$	0.265
Fe_{blood}	$y = -0.001x + 0.938$	0.12	$1.769_{1,13}$	0.206

Cu concentrations of both tail tissue (18.7 ± 35.7 μg/g) and blood (87.9 ± 43 μg/g) in lizards from mining site 1 site were greater than recorded Cu concentration values in tail tissue from *Psammodromus algirus* lizards occurring in an area affected by contaminants from a collapsed tailings dam in Spain (Márquez-Ferrando et al. 2009) (10.5 ± 1.6 μg/g in lizards from the contaminated site). The same was true for Zn concentrations in tail tissues compared with *P. algirus*, whereas Mn concentrations in tissue and blood were similar for both species (Márquez-Ferrando et al. 2009).

Contaminant concentrations recorded in blood from *S. giganteus* are generally greater than values reported in various tissues for other vertebrates collected at

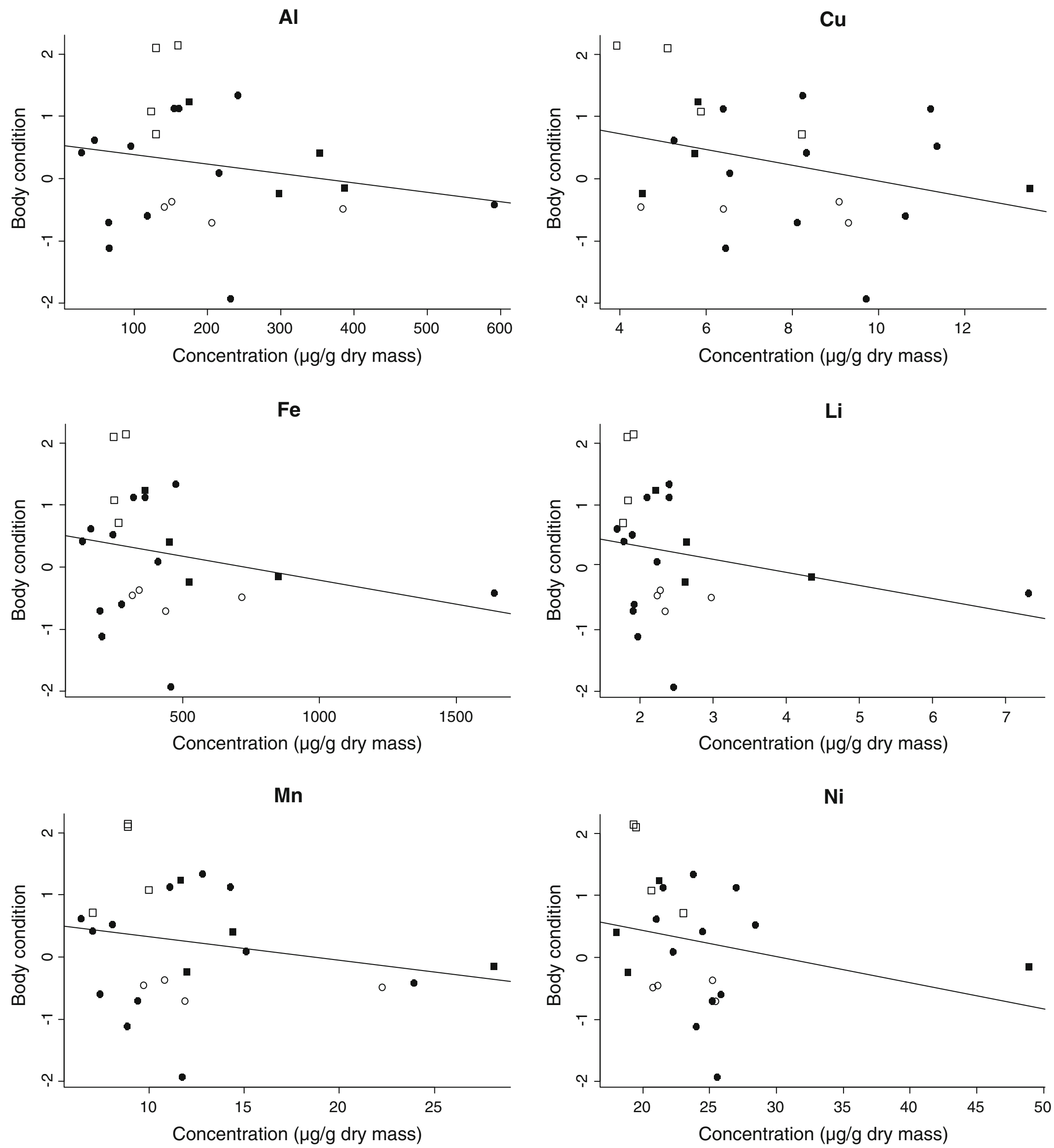

Fig. 4 Scatter plots indicating the negative relationships between some metal concentrations in tail tissue, and body condition of sampled lizards. *Filled circles* mining site 1; *filled blocks* mining site 2; *open blocks* control site 1; *open circles* control site 2

contaminated sites. For example, van Eeden and Schoonbee (1992) reported mean Ni concentrations of 40.9 μg/g dry mass in red-knobbed coot (*Fulica cristata*) organs sampled from birds at a wetland contaminated by seepage from mines and industry. The mean Ni concentration in sungazer lizard blood samples from mining site 1 was 5 times that value: 214 μg/g dry mass. In addition, van Eeden and Schoonbee (1992) reported mean Cr concentrations of 8.9 μg/g dry mass for the same bird species compared with 137.4 μg/g dry mass for *S. giganteus* blood samples from mining site 1. Fe and Mn concentrations for coots were, however, relatively similar to those of *S. giganteus* from mining site 1: coots = 1566 and 45.2 μg/g dry mass; *S. giganteus* = 1677.1 and 13.4 μg/g dry mass (blood samples), respectively.

Pathways of contaminant uptake were not investigated here, although we expect uptake to be primarily through diet, which has previously been documented for reptiles (e.g., Hopkins et al. 2002). However, sungazer lizards are burrowing animals, and the potential of direct uptake from contaminated soils cannot be excluded. Soil ingestion has also been shown to be a potentially important route for the uptake of contaminants in lizards (Rich and Talent 2009), especially because soil ingestion has been reported to occur under natural conditions in many reptile species

(Sokol 1971). Future investigations into the contaminant load of potential prey species at sites monitored here can likely clarify potential pathways of exposure by sungazer lizards.

Body Condition and Sex Ratio

None of the possibly toxic elements measured in tail tissue and blood sampled from *S. giganteus* showed statistically significant relationships with the condition of lizards sampled. Nonetheless, these relationships almost always exhibited a negative slope (Fig. 3; Table 2). We consider these relationships suggestive of negative effects on the body condition of lizards as a result of increased concentrations of a number of metals (particularly Al, Cu, Fe, Li, Mn, and Ni), especially given that mining site 1, our contaminated site, had the highest invertebrate abundance. Although the mechanisms for this effect are not understood, the poorer physiological condition of lizards from mining site 1 suggests the need to explore potential life history and reproductive costs associated with inorganic contaminants.

Hopkins et al. (2002) reported that high concentrations of six elements (As, Cd, Cu, Se, Sr, and V) accumulated through contaminated diet did not negatively affect food consumption, growth, condition factor, overwinter survival, metabolic rate, and gonadosomatic indices in banded water snakes (*Nerodia fasciata*). The potential influences of micronutrients on the toxic effects associated with increased metal concentrations (Peraza et al. 1998) may be partly responsible for the lack of statistically significant effects of increased metal concentrations on the body condition of lizards in our study as well as some other studies, such as Hopkins et al. (2002). Such influences are still poorly understood but may have substantial mitigating effects on the toxicity associated with nonessential metals. For instance, increased levels of Mg counteract many of the effects associated with increased uptake of Cd, thus decreasing Cd concentrations in kidney, spleen, and bone (Bulat et al. 2008) and also counteracting Cd-induced misbalances of bioelements (Bulat et al. 2012). The physiological consequences of the uptake of various combinations of inorganic compounds are therefore not always clear, and more detailed studies are needed.

The accumulation of metals in reptiles can be sex-specific due to differences in diet, physiology, and body size (Linder and Grillitsch 2000). For example, differences in metal concentrations between male and female *Anolis sagrei* sampled at various sites across southern Florida, USA, are attributed to differences in diet due to microhabitat differences in foraging locations (female lizards tending to feed closer to the ground than male lizards, which tend to feed on tree trunks and branches) (Burger et al. 2004). We found no significant differences in concentrations of all of the elements investigated between sexes of *S. giganteus*. This lack of differences in metal uptake between sexes is not surprising because no known feeding or other nonreproductive behavioural differences exist between sexes of this species (van Wyk 1992). Similarly, male and female *Psammodromus algirus* lizards from a pyrite mine tailings dam showed no differences in levels of inorganic contaminants (Márquez-Ferrando et al. 2009).

Mining site 1 was the only population to significantly depart from a 1:1 sex ratio. The two control sites consisted of 44 % male lizards compared with 29 % male lizards at mining site 1. Investigators van Wyk (1992) and Ruddock (2000) did not report any significant sex-based differences for feeding behaviour in *S. giganteus* that could potentially lead to greater levels of contaminant accumulation by either of the sexes. Furthermore, no significant differences were observed when concentrations of the various metals in blood and tail tissue were compared between sexes. Therefore, the reason for a female-biased sex ratio at mining site 1 is equivocal and could be due to chance or any number of factors that sometimes affect sex ratios.

In summary, the results of our study indicate that inorganic contaminants emanating from gold-mining effluents may have a negative impact on populations of *S. giganteus* in the Welkom area of the Free State Province of South Africa. Lizards from a mining-contaminated site had increased blood concentrations of Li, Na, Al, S, Si, Cr, Mn, Fe, Cu, Ni, W, and Bi. The effects of accumulation of these contaminants on lizards are poorly understood, but are related to a decrease in body condition, even in the presence of high prey availability. Further research into the mechanisms, effects and associated costs of mining-related contaminants on *S. giganteus* and lizards in general is needed to understand the fitness consequences and long-term population-level effects.

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